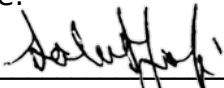


PARKINSON'S DISEASE THROUGH PERIPHERAL PROTEOMICS: CROSS-COHORT
DIAGNOSIS, SEVERITY MODELING, AND BIOLOGICAL GROUNDING FROM A
LEAKAGE-SAFE MULTI-OMICS FRAMEWORK

by

Nicholas Minster
A Dissertation
Submitted to the
Graduate Faculty
of
George Mason University
in Partial Fulfillment of
The Requirements for the Degree
of
Doctor of Philosophy
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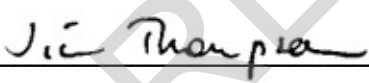
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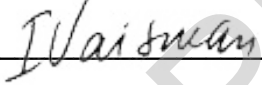
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DEDICATION

To my parents. They taught me the things that mattered and left the rest to me.

To my wife, Sara. You were steady when I was not. You never asked me to be different. I hope to do the same for you.

To my brothers who would have been better at this than me.

To my sisters-in-law Zahra and Leila, who made me part of the family and meant it.

And finally, to my mother-in-law and father-in-law:

مامان بابا شما قوی ترین آدم هایی هستید که میشناسم زمان هایی که خسته بودم و قوت ادامه نداشتم فکر کردن به شما به من امید و قوت ادامه کار داد. از صمیم قلب از شما متشکرم و دوستان دارم

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LIST OF ABBREVIATIONS AND SYMBOLS

ADL	Activities of Daily Living
AMI	Adjusted Mutual Information
AMP-PD	Accelerating Medicines Partnership Parkinson's Disease
ARI	Adjusted Rand Index
AUC	Area Under the Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
AUPRC	Area Under the Precision–Recall Curve
BH	Benjamini–Hochberg
BIC	Bayesian Information Criterion
BLUP	Best Linear Unbiased Predictor
CI	Confidence Interval
CLUE	Connectivity Map User Environment
CMap	Connectivity Map
CNS	Central Nervous System
CV	Cross-Validation
DCA	Decision Curve Analysis
ECM	Extracellular Matrix
EV	Extracellular Vesicle
FDR	False Discovery Rate
FLAIR	Fluid-Attenuated Inversion Recovery
GO	Gene Ontology
GSEA	Gene Set Enrichment Analysis
GMT	Gene Matrix Transposed
GWAS	Genome-Wide Association Study
HC	Healthy Control
HGNC	HUGO Gene Nomenclature Committee
HVG	Highly Variable Genes
ICA	Independent Component Analysis
IR3	Integrated Release 3
KEGG	Kyoto Encyclopedia of Genes and Genomes
LHS	Latin Hypercube Sampling
LM22	Leukocyte gene-expression signature matrix of 22 immune cell types
MAE	Mean Absolute Error
MAO-B	Monoamine Oxidase B
MDS-UPDRS	Movement Disorder Society Unified Parkinson's Disease Rating Scale
MHC	Major Histocompatibility Complex
MNI152	Montreal Neurological Institute 152 template
M0	Month 0 (baseline visit)
MRI	Magnetic Resonance Imaging
MSI	Molecular Severity Index

NGS.....	Next-Generation Sequencing
NNLS	Non-Negative Least Squares
NPX.....	Normalized Protein Expression
OLS	Ordinary Least Squares
OOF.....	Out-of-Fold
PC.....	Principal Component
PCA.....	Principal Component Analysis
PD	Parkinson's disease
PDBP.....	Parkinson's Disease Biomarkers Program
PEA	Proximity Extension Assay
pQTL.....	protein Quantitative Trait Locus
PPI.....	Protein-Protein Interaction
PPMI	Parkinson's Progression Markers Initiative
PSI.....	Proteomic Severity Index
PWAS	Proteome-Wide Association Study
q.....	Adjusted p-value (FDR-corrected)
QC	Quality Control
RNA-seq	RNA Sequencing
R ²	Coefficient of Determination
ROC	Receiver Operating Characteristic
RMSE.....	Root Mean Squared Error
SE.....	Standard Error
SHAP	SHapley Additive exPlanations
SN	Substantia Nigra
SPECT.....	Single Photon Emission Computed Tomography
STRING	Search Tool for the Retrieval of Interacting Genes/Proteins
SVD.....	Singular Value Decomposition
T1	T1-weighted MRI
T2	T2-weighted MRI
TEST	Held-out test set
TLR.....	Toll-like receptor
TPM	Transcripts Per Million
TRAIN	Training set
UPDRS.....	Unified Parkinson's Disease Rating Scale
UPSIT	University of Pennsylvania Smell Identification Test
WGCNA	Weighted Gene Co-expression Network Analysis
ρ_s	Spearman rank correlation coefficient

DISSERTATION ABSTRACT

PARKINSON'S DISEASE THROUGH PERIPHERAL PROTEOMICS: CROSS-COHORT DIAGNOSIS, SEVERITY MODELING, AND BIOLOGICAL GROUNDING FROM A LEAKAGE-SAFE MULTI-OMICS FRAMEWORK

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George Mason University, 2026

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Parkinson's disease (PD) affects over ten million people worldwide, yet diagnosis remains clinical and occurs after substantial neurodegeneration. This dissertation develops an interpretable, leakage-safe machine learning framework for PD biomarker discovery using matched whole-blood RNA sequencing and targeted plasma proteomics from the Accelerating Medicines Partnership–Parkinson's Disease (AMP-PD), with PPMI and PDBP as training and external validation cohorts.

The first study trains a proteomics-only Random Forest classifier that achieves external AUROC = 0.8724 (95% CI 0.8305–0.9097). Whole-blood RNA yields modest discrimination (AUROC = 0.5978), and no multimodal fusion strategy significantly improves on proteomics alone. Recursive feature elimination identifies a 32-protein signature enriched for extracellular signaling and immune pathways. A Proteomic Severity

Index explains 28.2% of baseline MDS-UPDRS variance but does not independently predict progression after baseline adjustment.

The second study reports negative results from unsupervised subtyping. Consensus clustering of residualized whole-blood RNA produces highly stable partitions that lack clinical separation or PD-specific biology. Plasma proteomic associations with progression do not survive multiple-testing correction. These results demonstrate that cluster stability alone is insufficient for biological interpretability and that discrete molecular subtypes are not supported by this dataset.

The third study externally validates a continuous severity model. Ridge regression on 1,168 plasma proteins generalizes across cohorts (participant-mean Spearman $\rho = 0.507$; 95% CI 0.366–0.628), with confirmatory mixed-effects analyses replicating protein–severity associations and demonstrating incremental prognostic signal beyond baseline UPDRS. Enrichment converges on cell–matrix adhesion and integrin-mediated signaling, with genetic anchoring identifying overlap with PD risk loci.

A translational chapter extends these findings through proteomics–MRI integration identifying TNF-enriched co-expression modules with subcortical volumetric correspondence, Connectivity Map analysis nominating therapeutic hypotheses including an mGluR4 modulator, and Mendelian randomization feasibility demonstrating strong cis-pQTL instruments for ten severity-associated proteins. Across all studies, plasma proteomics consistently outperforms transcriptomics under cross-cohort transfer, with biological themes converging on extracellular signaling, immune regulation, and cell–matrix adhesion.

CHAPTER 1: INTRODUCTION

Parkinson's disease (PD) is a progressive neurodegenerative disorder marked by dopaminergic neuron loss and the accumulation of α -synuclein based Lewy bodies [1]. Clinical manifestations include a range of progressive motor and non-motor symptoms that erode quality of life and increase healthcare burden worldwide. Over ten million people are affected globally [2]. Despite this large burden, diagnosis remains primarily qualitative and occurs only after substantial neurodegeneration, limiting the window or barring disease modifying intervention [3, 4]. Current therapies such as levodopa and exercise address symptoms but do not alter the underlying disease course, during which time adverse effects can accumulate [5-7]. There is a need for biologically anchored methods that enable early detection and personalized care [8].

High-throughput molecular profiling and the release of large, PD-specific cohorts now create an opportunity to move beyond a symptom-based paradigm [9, 10]. Blood-based proteomics and RNA-sequencing offer scalable, minimally invasive records of disease state [11, 12]. Yet, the analytical pace has lagged data generation, and many datasets remain insufficiently explored [13]. This dissertation addresses noted gaps in the literature by integrating proteomic and transcriptomic signals through interpretable, leakage safe machine learning to deliver validated biomarkers, molecular subtypes, and mechanistic insights that can translate to clinical decision-making and trial design.

PROBLEM STATEMENT

Two linked gaps impede progress. First, there is no widely adopted, noninvasive biomarker set that reliably detects prodromal or early PD or stratifies heterogeneous patients, that generalizes across cohorts [14]. Second, methodological limitations such as, fragmented and poor modality selection, inadequate controls for data leakage, limited external validation, and limited model interpretation constrain reproducibility and clinical translation [15]. Rigorous, multimodal, and interpretable frameworks are needed to extract biologically relevant knowledge.

LITERATURE SYNTHESIS

PD biology centers on α -synuclein aggregation, mitochondrial dysfunction, neuroinflammation, and autophagic-lysosomal impairment. Monogenic forms (e.g. GBA1, LRRK2) highlight that many disease pathways are convergent and may inform idiopathic disease [16]. This dissertation focuses on idiopathic PD, which makes up approximately 90% of cases [17]. Development of diagnostics aims to move beyond qualitative measures, α -synuclein seeding assays show promise in selected subgroups but are unlikely to change the standard of care [18]. Neuroimaging increasingly detects early structural and functional changes [19]. Wearables and other technology can successfully detect slight changes in motor and non-motor function but have limited impact on informing biology. Peripheral biomarkers, like α -synuclein species, neurofilaments, and inflammatory mediators have gained attention due to accessibility but require rigorous validation and integration to inform biology and reach clinical practice. These biomarkers may also inform progression or subtypes to inform personalized medicine [14, 20].

CONCEPTUAL FRAMEWORK

This dissertation operationalizes a data-to-translation pipeline: (i) derive robust, leakage-safe supervised models from blood proteomics and transcriptomics to predict diagnosis and quantify disease severity; (ii) map model features to pathways, ontologies, and clinical/imaging correlates; (iii) evaluate longitudinal dynamics and progression relevance using mixed-effects frameworks and strict external validation; and (iv) test whether peripheral molecular subtypes are supported under stability and transfer criteria, versus whether heterogeneity is better represented as a continuous severity axis with decision-analytic utility.

The premise of this framework is that interpretable machine learning on peripheral molecular data can detect PD, estimate severity, nominate mechanistic programs and correlates, and support clinically useful stratification under leakage-safe cross-cohort evaluation.

RESEARCH AIMS AND OBJECTIVES

Aim 1. Diagnostic and severity biomarkers. Develop leakage-safe, interpretable models from harmonized blood proteomic and transcriptomic data to predict PD diagnosis and quantify motor severity. Evaluate performance, calibration, and error using participant-grouped cross-validation, then confirm generalization in an external holdout cohort.

Aim 2. Biological and clinical grounding. Ground predictive features in PD-relevant biology by mapping them to pathways and ontologies (GO/KEGG/Reactome), cellular context, and interaction-network modules. Test associations with clinical endpoints (for example, UPDRS and UPSIT) using multiple-testing-controlled correlation

and enrichment analyses, including ORA, ranked enrichment, and network propagation for module coherence.

Aim 3. Dynamics and therapeutic hypotheses. Quantify longitudinal biomarker dynamics and progression relevance using mixed-effects models to estimate disease-direction signals over time. Prioritize candidate interventions by scoring drug-perturbation signatures against direction-specific molecular signals to generate testable therapeutic hypotheses.

Aim 4. Stratification. Evaluate whether peripheral omics supports robust, transferable discrete subtypes under stability-aware clustering and strict external validation. If discrete subtypes are not supported, operationalize heterogeneity as an externally validated continuous severity axis and clinically interpretable strata (for example, thresholds and quantiles), then quantify stratification utility using effect sizes and decision-analytic metrics.

This dissertation advances AMP-PD goals by translating harmonized multi-omic resources into leakage-safe, clinically interpretable outputs. Building on prior work demonstrating externally validated blood-based classifiers, the dissertation develops calibrated, interpretable biomarkers and transportable models that move beyond discrimination toward mechanistic insights and translational relevance [21]. In parallel with the broader literature showing that clinically grounded longitudinal phenotyping can recover reproducible progression patterns, this work tests whether peripheral omics supports transferable stratification across cohorts and if it connects robust molecular

signals to pathway programs, imaging correlates, and perturbational signatures to motivate therapeutic hypotheses or trial enrichment [22, 23].

This dissertation follows a manuscript-style format with three research chapters. Each chapter opens with a short Literature Review that serves as a preface, not part of the corresponding manuscript. This is intended to orient the reader to scope, datasets, and prior work. References in these sections are in the general references.

PREVIEW

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